

Python

Python is a popular programming language. It was created by Guido van Rossum, and released in 1991.

Applications

- Python can be used on a server to create web applications.
- Python can be used alongside software to create workflows.
- Python can connect to database systems. It can also read and modify files.
- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid prototyping, or for production-ready software development.
- **Web Development**
 - Framework such as **Django, Pyramid**
 - Micro-Framework such as **Flask, Bottle**
- **Desktop-based Applications**
 - Image, Video, Audio Processing (using modules like **OpenCV** and **PyAudio** or **librosa**)
 - Graphic Design Applications (used to create **Inkscape, GIMP, Blender, 3ds Max** softwares)
 - Games (**PySoy** -> 3D game engine, **PyGame** -> library for game development)
- **Scientific and Computational Applications**
 - Machine Learning (Regression, Decision Tree, Random Forest, Deep Learning) using libraries such as **NumPy, SciPy, TensorFlow, Keras, PyTorch**
- **Internet Protocol**
 - HTML, XML
 - JSON
 - Requests
 - BeautifulSoup

Why Python?

It is a general-purpose, high-level, interpreted, dynamic programming language.

1. General-Purpose

- designed to be used for writing software in the widest variety of application domains

1. Simple Syntax

- readability
- Python syntax is similar to the English language
- Python syntax allows developers to write programs with fewer lines than some other programming languages.
- Python relies on indentation, using whitespace, to define scope; such as the scope of loops, functions and classes. Other programming languages often use curly-brackets for this purpose.

1. Open Source

- We can use python software without any license and its freeware.

1. Portability

- designed to be more or less independent of a particular type of computer, human-readable friendly
- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc)

1. Interpreted

- designed to execute instructions of a program directly, without previously compiling a program into machine-language instructions.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick.

1. High-level

- Programmer friendly, we not required to concentrate low level activities like memory management and security etc
- Python can be treated in a procedural way, an object-oriented way or a functional way.

7. Dynamic Type

- we no need assign datatype while assigning variable

1. Robust Standard Library

- Python supports modules and packages, which encourages program modularity and code reuse.

But, what is the difference between an interpreter and a compiler?

Interpreter	Compiler
Executes program by taking one statement at a time	Translates the entire program at once into machine code

Interpreter	Compiler
Error is reported as soon as the first error is encountered . Won't show the next set of errors if the existing one isn't solved	Errors are reported after the entire program is checked
Easier to debug	Difficult
No intermediate object code is generated, hence memory efficient	Generates intermediate object code which further requires linking, hence requires more memory
Overall execution is slower	Faster

Python can be executed in various ways

Without IDE

1. Writing a code in notepad and execute in command prompt
2. Python (command prompt)
3. Python IDLE
4. Run file in Python IDLE

With IDE (integrated Development Environment)

1. Jupyter notebook
2. Spyder
3. Pycharm, Netbeans or Eclipse which are particularly useful when managing larger collections of Python files.